

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMOE505D - PRODUCT DESIGN ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The selection of materials for a product is mostly determined by:-----
 - A) Appearance only
 - B) Availability in market
 - C) Functional requirements and cost
 - D) Designer preference
- 2 The primary applications of reverse engineering are:-----
 - A) Product improvement and redesign
 - B) Patent filing
 - C) Marketing analysis
 - D) Inventory control
- 3 In product design, ergonomics emphasizes:-----
 - A) Manufacturing speed
 - B) Material strength
 - C) Cost reduction
 - D) User comfort and safety
- 4 The process of generating a large number of ideas in a short time is called-----
 - A) Benchmarking
 - B) Brainstorming
 - C) Reverse engineering
 - D) Prototyping
- 5 Product architecture defines:
 - A) External appearance of product
 - B) Arrangement of functional elements and components
 - C) Product cost
 - D) Market strategy

- 6 Design automation primarily helps in:
 - A) Manual drafting
 - B) Eliminating testing
 - C) Increasing design complexity
 - D) Reducing design time and errors

- 7 Which phase benefits most from prototype feedback?
 - A) Conceptual design
 - B) Marketing
 - C) Packaging
 - D) Advertising

- 8 Which tool is commonly used for decision making in manufacturing?
 - A) Brainstorming
 - B) Assembly drawing
 - C) Decision matrix
 - D) Prototype

- 9 Which document provides instructions for production and inspection?
 - A) Market survey report
 - B) Advertisement brochure
 - C) Sales invoice
 - D) Engineering drawing

- 10 Automation is most suitable for:
 - A) Job production
 - B) Mass production
 - C) Repair work
 - D) Experimental models

- 11 Reuse of products helps mainly in:
 - A) Increasing production cost
 - B) Increasing waste generation
 - C) Conserving resources and energy
 - D) Reducing product life

- 12 Designing products with recyclable materials mainly reduces:
 - A) Environmental pollution
 - B) Product quality
 - C) Customer satisfaction
 - D) Production speed

Q2. Solve the following.

- A) Explain the systematic problem-solving approach used in product design. 6
Illustrate the steps with a neat flow diagram.
- B) Describe how product disassembly and reverse engineering aid in product redesign and improvement. 6

Q3. Solve the following.

- A) Describe the techniques used by both individuals and groups to shortlist ideas for product design. 6
- B) Discuss how component design incorporates Design for Manufacturability (DFM). 6

- Q4. Solve Any Two of the following.
- A) Describe various types of prototypes that are used in product design and discuss their applications. 6
 - B) Explain the importance of process safety in product design and manufacturing. 6
 - C) Explain the concept of multidisciplinary teams in product design. Discuss their advantages and challenges. 6
- Q5. Solve Any Two of the following.
- A) Define Material Safety Data Sheet (MSDS). Explain its contents and importance in manufacturing industries. 6
 - B) Explain the factors considered in selecting a manufacturing process for a given product. 6
 - C) With a suitable example, explain how CAD/CAM, decision models, MSDS, and manufacturing scale are integrated in modern manufacturing. 6
- Q6. Solve Any Two of the following.
- A) Define Product Life-Cycle Management (PLM) and discuss the different stages of a product life cycle. 6
 - B) Talk about how Life Cycle Assessment (LCA) is used to assess a product's environmental performance. 6
 - C) Discuss the advantages and limitations of recycling engineering products with suitable examples. 6

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